A Multivariate Scan Statistic for Spatial Variation in Nonlinear Temporal Trends: Application to Gonorrhea and Chlamydia in the USA

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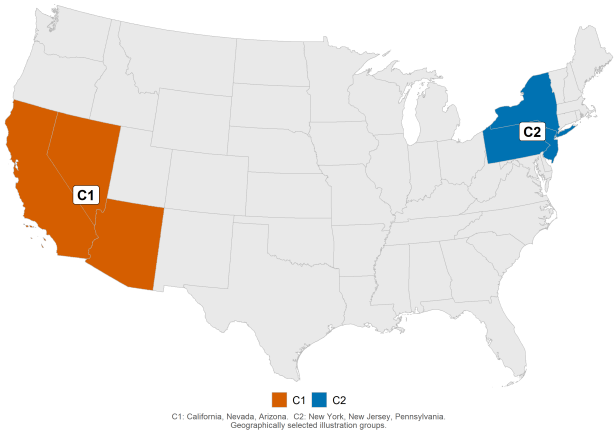
November 3, 2026

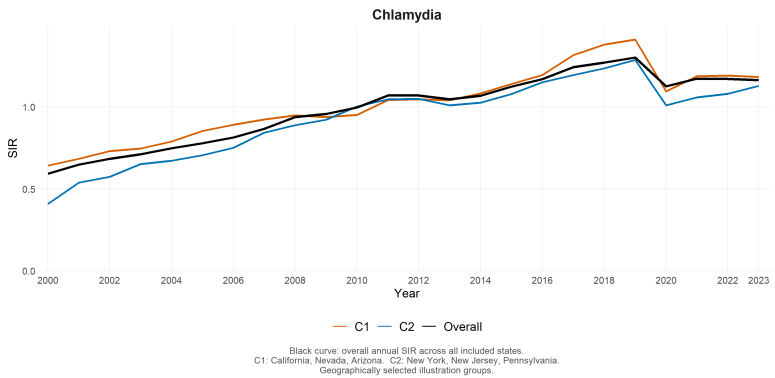
Introduction

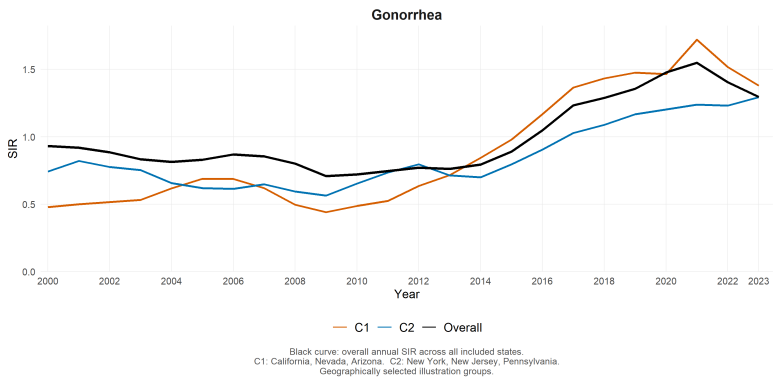
Spatial Clusters and Temporal Trends

Detecting geographic variations in disease risk temporal trends is important for decision-making in public health surveillance. Trends may differ in direction and shape, and involve multiple diseases.

Objective: Identify spatial clusters where the risk of multiple diseases differ significantly from the trends outside the cluster.







What Comes Next



1. **Existing Methods**
   HD Spatial

2. **Understanding MSVTT**
   Model, hypotheses, joint scan statistic, and significance testing

3. **Simulation study**
   Simulation design and performance evaluation

4. **Application to gonorrhea and chlamydia in the USA**
   Spatial clusters and disease-specific temporal trends

5. **Conclusion**

Existing Methods

HDSpatialScan: Multivariate Methods

Multiple disease curves are observed at each location.

Method	**Working**
MPFSS-LH	Compares mean vector-valued curves inside and outside candidate regions using functional MANOVA and the Lawley–Hotelling trace.
MDFFSS	Compares disease vectors inside and outside candidate regions using pointwise Hotelling T^2 statistics.
MRBFSS	Uses pointwise multivariate Wilcoxon rank-sum statistics to compare observations inside and outside candidate regions.
NPFSS	Compares entire vector-valued curves using a functional Wilcoxon–Mann–Whitney statistic.

[1]

Limitations for Disease-Count Applications



Functional scan methods: MPFSS-LH, MDFFSS, MRBFSS, and NPFSS.

- **No explicit disease-count likelihood.** Although SIRs standardize observed counts by expected counts, these methods analyse the resulting curves without explicitly specifying a count likelihood with expected-count offsets.
- **Separating baseline risk from temporal change.** Curve differences may reflect persistent differences in risk levels as well as temporal patterns. These methods do not directly isolate temporal-trend differences while allowing location-specific baseline risks.
- **Exchangeability of SIR curves.** Spatial permutation requires exchangeability across locations under the null. SIR variability can differ between counties with different expected counts, so SIR standardization alone does not guarantee this condition.

Proposed Method: Multivariate SVTT



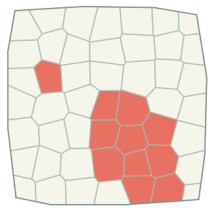
- We propose MSVTT, a flexible spatial scan method to detect spatial variations of multiple disease risk temporal trends
- MSVTT is a SVTT-based method that detects clusters whose disease-specific temporal trends differ from those outside the region
- The method models disease counts using Poisson regression with population-based offsets. It uses splines functions to capture nonlinear patterns separately for each disease
- The method compares shared versus separate inside–outside trends using sum disease-specific likelihood-ratio statistics. Significance is assessed using Monte Carlo simulation

Proposed MSVTT method

Spatial Scan Statistics: The Basic Idea

Find geographic regions with unusually high disease risk.

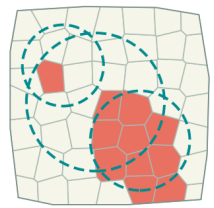
1 Observe the data



Cases + population
for each area

Higher observed rate

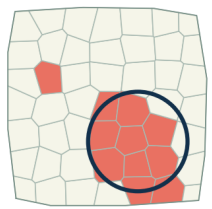
2 Scan candidate regions



Move and resize
a scanning window.

Compare risk inside
with risk outside.

3 Select the strongest signal

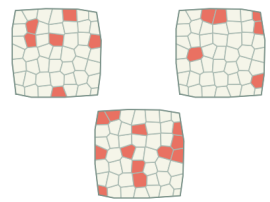


Calculate a score
for every window.

Keep the window with
the largest score.

4 Test against chance

No-cluster simulations



Simulate under no cluster
and repeat the full scan.

Compare maximum scores
to obtain a p-value.

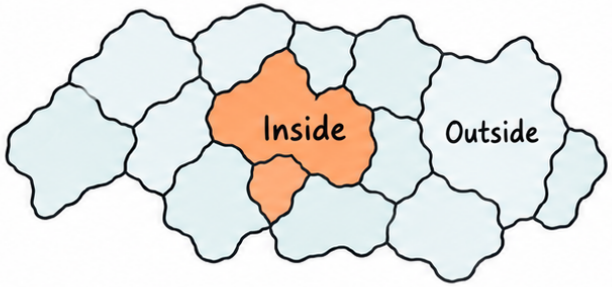
Scan statistic = largest window score. A small p-value provides evidence of a cluster.

Schematic illustration. Circular windows are one common choice; other shapes can also be used.

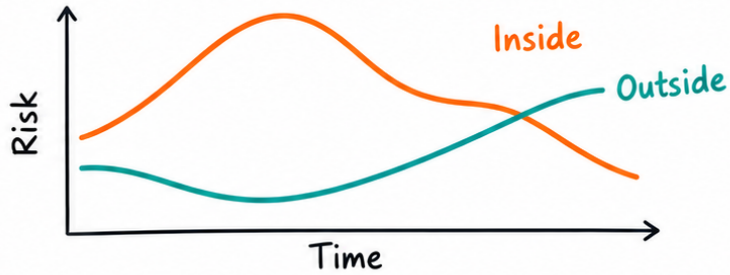
SVTT

Finding regions with different time trends

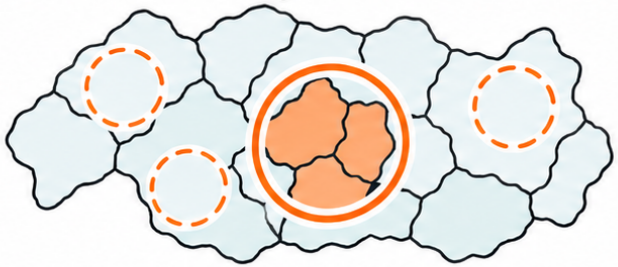
1. Choose a region



2. Compare time trends

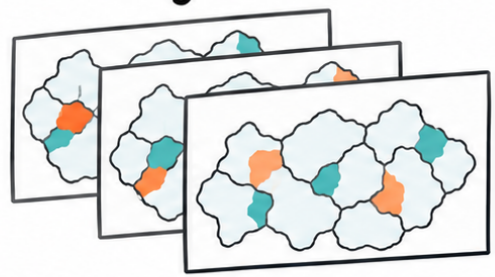


3. Scan all regions



Find the largest likelihood ratio

4. Test significance



Compare with null simulations

MVSVTT: Data and Assumptions



Data and notation

- D: number of diseases, $d = 1, \ldots, D$.
- $i = 1, \ldots, n$: spatial units.
- $t = 1, \ldots, T$: observation times.
- Y_{idt}: observed cases.
- P_{idt}: population or exposure at risk.

Disease-specific pooled baseline

$$r_{0d} = \frac{\sum_{i,t} Y_{idt}}{\sum_{i,t} P_{idt}},$$

$$m_{idt} = P_{idt} r_{0d} > 0.$$

Each disease has its own reference rate, pooled over space and time.

Candidate spatial cluster

For $Z \in \mathcal{Z}$, let $g \in \{Z, Z^c\}$.

$$Y_{dgt} = \sum_{i \in g} Y_{idt},$$

$$m_{dgt} = \sum_{i \in g} m_{idt}.$$

Assumptions

- Candidate regions are prespecified, with Z and Z^c nonempty.
- Counts are conditionally independent Poisson variables.
- Diseases are conditionally independent, with separate parameters.
- Positive expected counts serve as fixed offsets.

MSVTT: Model and Hypotheses

For each disease d, group $g \in \{Z, Z^c\}$, and time t:

$$Y_{dgt} \sim \text{Poisson}(\mu_{dgt}),$$
$$\log \mu_{dgt} = \underbrace{\log m_{dgt}}_{\text{offset}} + \alpha_{dg} + \underbrace{\mathbf{b}(t)^\top \boldsymbol{\beta}_{dg}}_{f_{dg}(t)}.$$

Flexible temporal basis

$$f_{dg}(t) = \sum_{k=1}^{K} \beta_{dgk} b_k(t).$$

- K: basis dimension, using polynomial or restricted cubic spline terms.
- Fix the basis and knots before scanning, for all windows and both models.
- Full-rank columns with $b_k(t_0) = 0$.

Multivariate trend comparison

$$\mathbf{f}_g(t) = \left(f_{1g}(t), \ldots, f_{Dg}(t)\right)^\top.$$

$$H_0(Z): \quad \boldsymbol{\beta}_{dZ} = \boldsymbol{\beta}_{dZ^c}, \ \forall d,$$
$$H_1(Z): \quad \boldsymbol{\beta}_{dZ} \neq \boldsymbol{\beta}_{dZ^c}, \ \exists d.$$

Thus H_0 means $\mathbf{f}_Z(t) = \mathbf{f}_{Z^c}(t)$ at every observed time.
The alternative allows differences in one or more diseases.

Separate intercepts: α_{dZ} and α_{dZ^c} remain free under both models, allowing baseline risk differences while testing temporal trends.

MSVTT: Likelihood and Model Fitting



Poisson log-likelihood for disease d and candidate Z

$$\ell_d(\boldsymbol{\theta}_d; Z) = \sum_{g \in \{Z, Z^c\}} \sum_{t=1}^{T} \left[Y_{dgt} \log \mu_{dgt} - \mu_{dgt} - \log\left(Y_{dgt}!\right) \right].$$

Maximum-likelihood fitting for each disease

- H_0 ($h = 0$): shared trend coefficients ($K + 2$ parameters).
- H_1 ($h = 1$): separate trend coefficients ($2K + 2$ parameters).

Both models retain two separate intercepts. $\boldsymbol{\theta}_d$ contains the intercepts and trend coefficients.

$$\widehat{\ell}_{d,h}(Z) = \max_{\boldsymbol{\theta}_d \in \Theta_{d,h}} \ell_d(\boldsymbol{\theta}_d; Z), \qquad h \in \{0, 1\}.$$

Disease-specific and joint evidence

$$\mathrm{LLR}_d(Z) = \widehat{\ell}_{d,1}(Z) - \widehat{\ell}_{d,0}(Z).$$

Under conditional independence across diseases:

MSVTT: Spatial Scan and Significance



Observed scan and most likely cluster

$$S_{\text{joint}}^{\text{obs}} = \max_{Z \in \mathcal{Z}} \text{LLR}_{\text{joint}}(Z),$$

$$Z^* \in \arg\max_{Z \in \mathcal{Z}} \text{LLR}_{\text{joint}}(Z).$$

Fitted global null for disease d

$$\log \mu_{idt}^{(0)} = \log m_{idt} + \gamma_{id} + f_{0d}(t).$$

- γ_{id}: baseline log-risk for spatial unit i and disease d.
- $f_{0d}(t)$: common temporal trend across all units for disease d.
- Fit this null by maximum likelihood before simulation.

Monte Carlo calibration

1. Generate B complete datasets under the fitted global null.
2. Repeat the entire scan, using the same windows, basis, and fitting procedure.

$$S_{\text{joint}}^{(b)} = \max_{Z \in \mathcal{Z}} \text{LLR}_{\text{joint}}^{(b)}(Z).$$

$$\widehat{p} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left\{ S_{\text{joint}}^{(b)} \geq S_{\text{joint}}^{\text{obs}} \right\}}{B + 1}.$$

The observed maximum is significant when $\widehat{p} \leq \alpha$, where α is the chosen significance level.

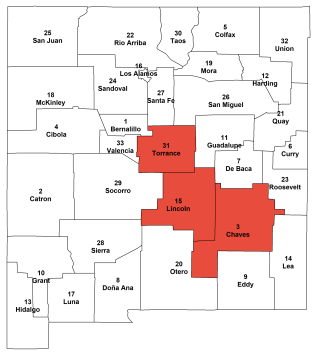
Dependent diseases: The summed marginal log-likelihood ratios form a composite-likelihood scan statistic. Null simulations must account for cross-disease dependence.

Simulation Study

Simulation Study: Setting

We evaluate the performance of the proposed MVSVTT method in comparison with existing approaches.

- Consider two diseases. Spatial domain is the 33 counties of New Mexico, United States. Temporal domain is 21 time points
- Consider a number of cluster specifications and disease risk scenarios
- Simulate disease counts over time inside and outside the clusters
- Fit MSVTT and existing approaches and obtain performance measures



Simulation Study: Data Generation



- **Count model:**

$$Y_{itd} \sim \text{Poisson}(m_{itd} R_{itd}), \qquad m_{itd} = 0.01 N_i,$$

where N_i is the population of county i.

- **Scenario-specific relative risks:**

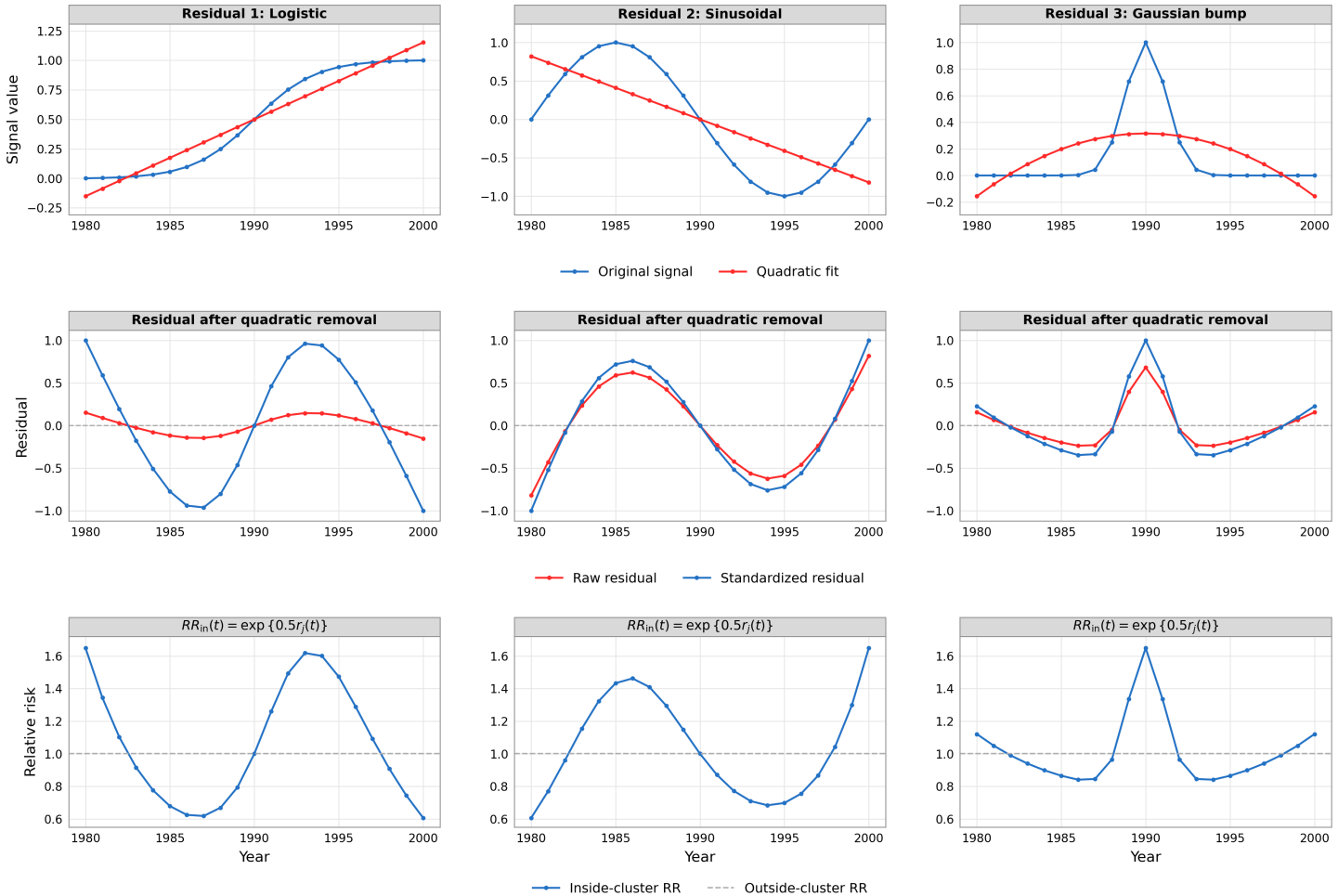
$$R_{itd} = \begin{cases} \exp\{\delta r_d(t)\}, & i \in Z_0, \\ 1, & i \notin Z_0, \end{cases} \qquad \delta = 0.1, 0.2, 0.3, 0.5.$$

Four scenarios specify the nonlinear trajectories $r_d(t)$.

- **Independence:** Counts generated independently across counties, time points, and diseases.

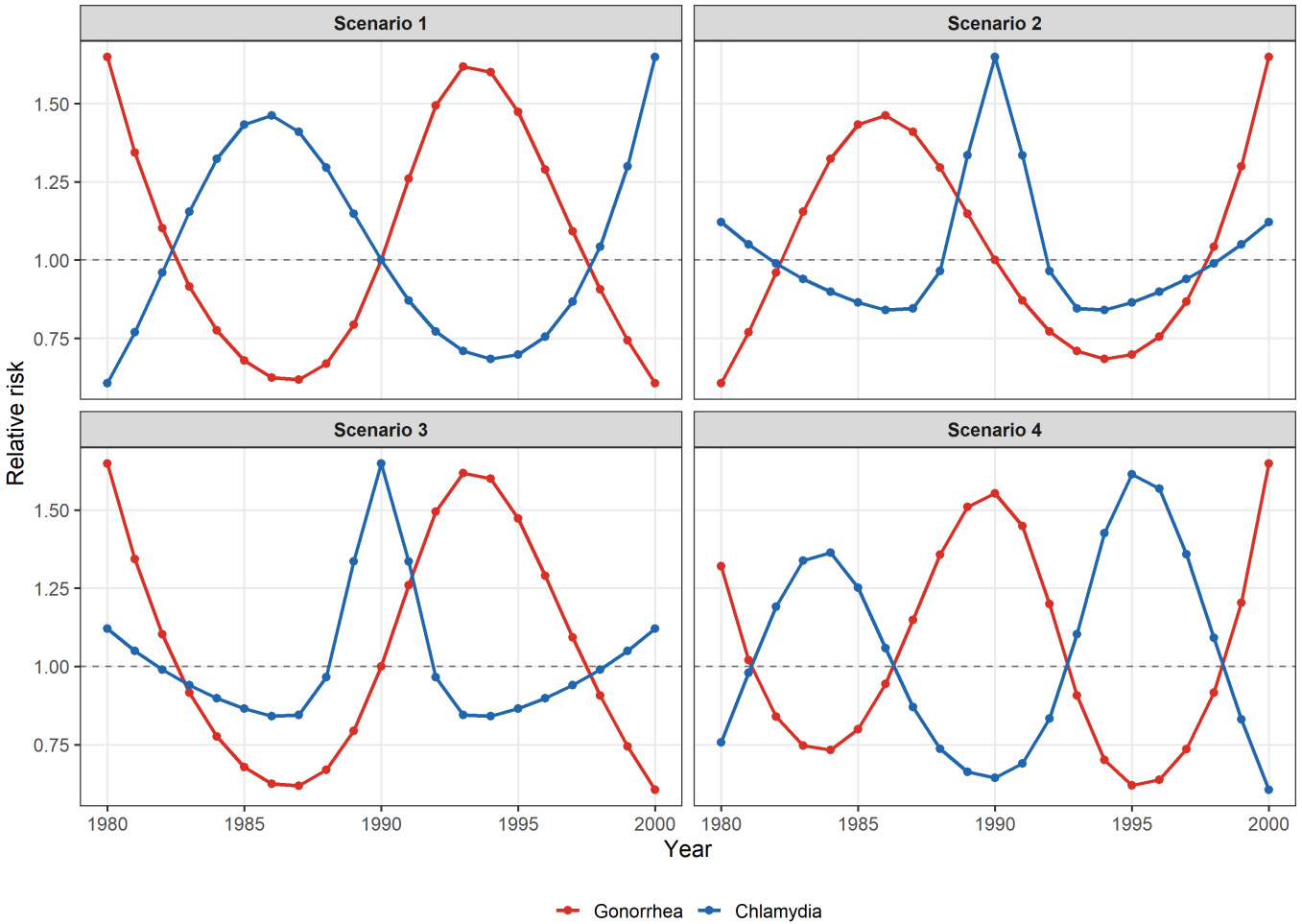
Simulation Study: Temporal Risk Profiles

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Simulation Study: Relative Risk Scenarios

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Simulation performance: 100 replicates



County-specific populations, $\delta = 0.1$

Scenario	Method	Power	Sensitivity	Specificity	Precision	Jaccard
S1	MPFSS-LH	0.0100	0.0000	**0.9997**	0.0000	0.0000
	MDFFSS	0.0300	0.0000	0.9980	0.0000	0.0000
	MRBFSS	0.1400	0.0367	0.9697	0.0148	0.0145
	NPFSS	0.6700	0.6300	0.9910	0.5875	0.5875
	MVSVTT	**1.0000**	**1.0000**	0.9950	**0.9630**	**0.9630**
S2	MPFSS-LH	0.0300	0.0000	0.9990	0.0000	0.0000
	MDFFSS	0.0100	0.0000	**0.9997**	0.0000	0.0000
	MRBFSS	0.1500	0.0200	0.9667	0.0075	0.0070
	NPFSS	0.2000	0.1500	0.9910	0.1250	0.1250
	MSVTT	**1.0000**	**0.9930**	0.9870	**0.9010**	**0.8940**
S3	MPFSS-LH	0.0100	0.0000	**0.9997**	0.0000	0.0000
	MDFFSS	0.0200	0.0000	0.9990	0.0000	0.0000
	MRBFSS	0.1400	0.0467	0.9700	0.0195	0.0179
	NPFSS	0.4100	0.3533	0.9867	0.3058	0.3052
	MSVTT	**1.0000**	**1.0000**	0.9940	**0.9550**	**0.9550**
S4	MPFSS-LH	0.0400	0.0000	**0.9987**	0.0000	0.0000
	MDFFSS	0.0500	0.0000	0.9977	0.0000	0.0000
	MRBFSS	0.1500	0.0300	0.9690	0.0118	0.0110
	NPFSS	0.6300	0.6200	0.9913	0.5688	0.5688
	MSVTT	**1.0000**	**1.0000**	0.9940	**0.9550**	**0.9550**

S1: Res1–Res2; S2: Res2–Res3; S3: Res1–Res3; S4: Opposite.
Bold: best value within each scenario, including ties.

Geographical variations of temporal trends of gonorrhea and chlamydia in the United States

Joint Disease Surveillance

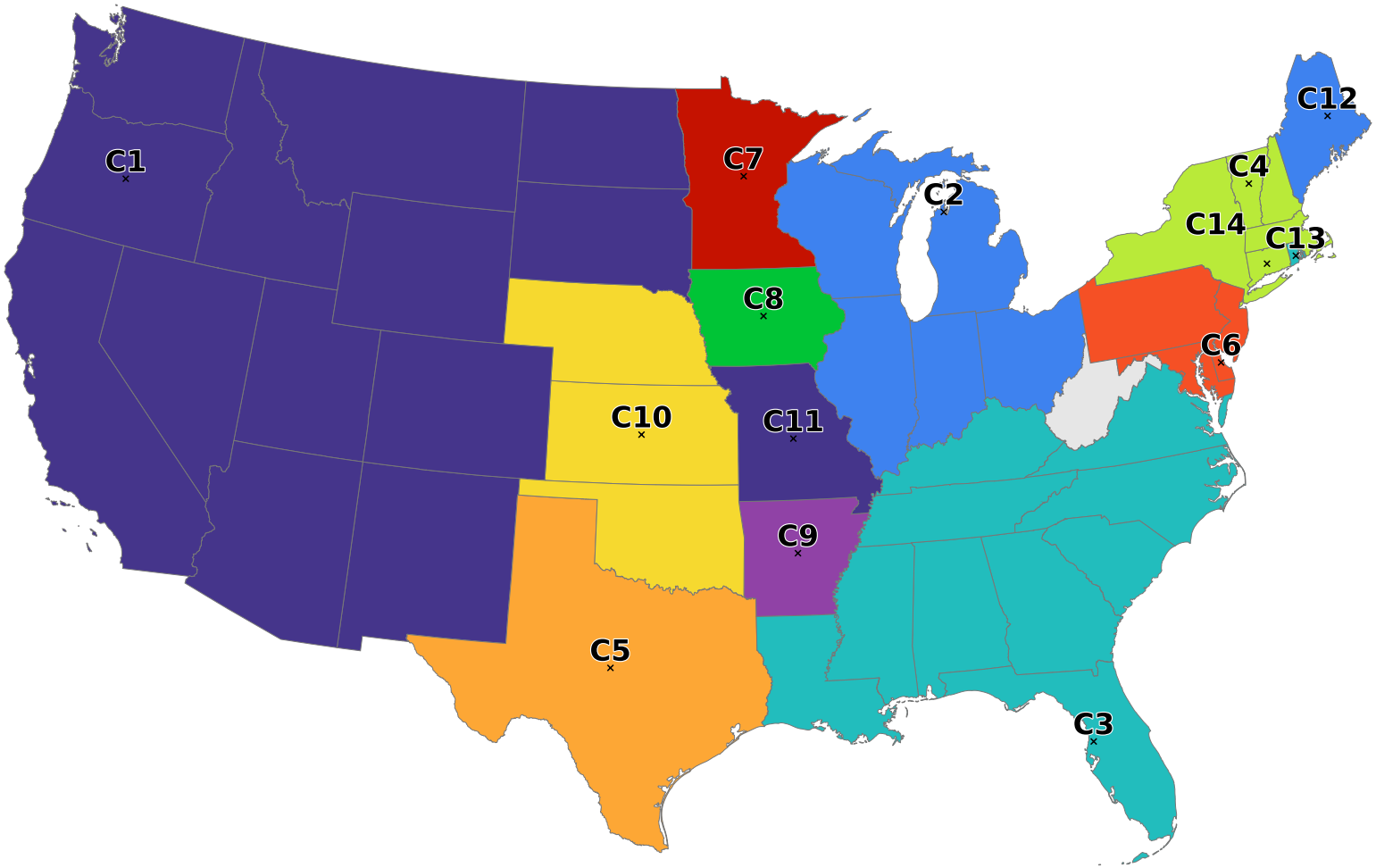


In 2023, CDC recorded **1,648,568 chlamydia cases** and **601,319 gonorrhea cases** in the United States [2].

- **National surveillance:** both infections are nationally notifiable, with reported cases available by location and year [2].
- **Interpreting temporal trends:** reported cases reflect infection incidence, screening coverage, diagnostic methods, and reporting practices [2].
- **Our multivariate application:** each state contributes two disease trajectories ($D = 2$). MSVTT identifies regions where trends in one or both diseases differ from those outside the region.

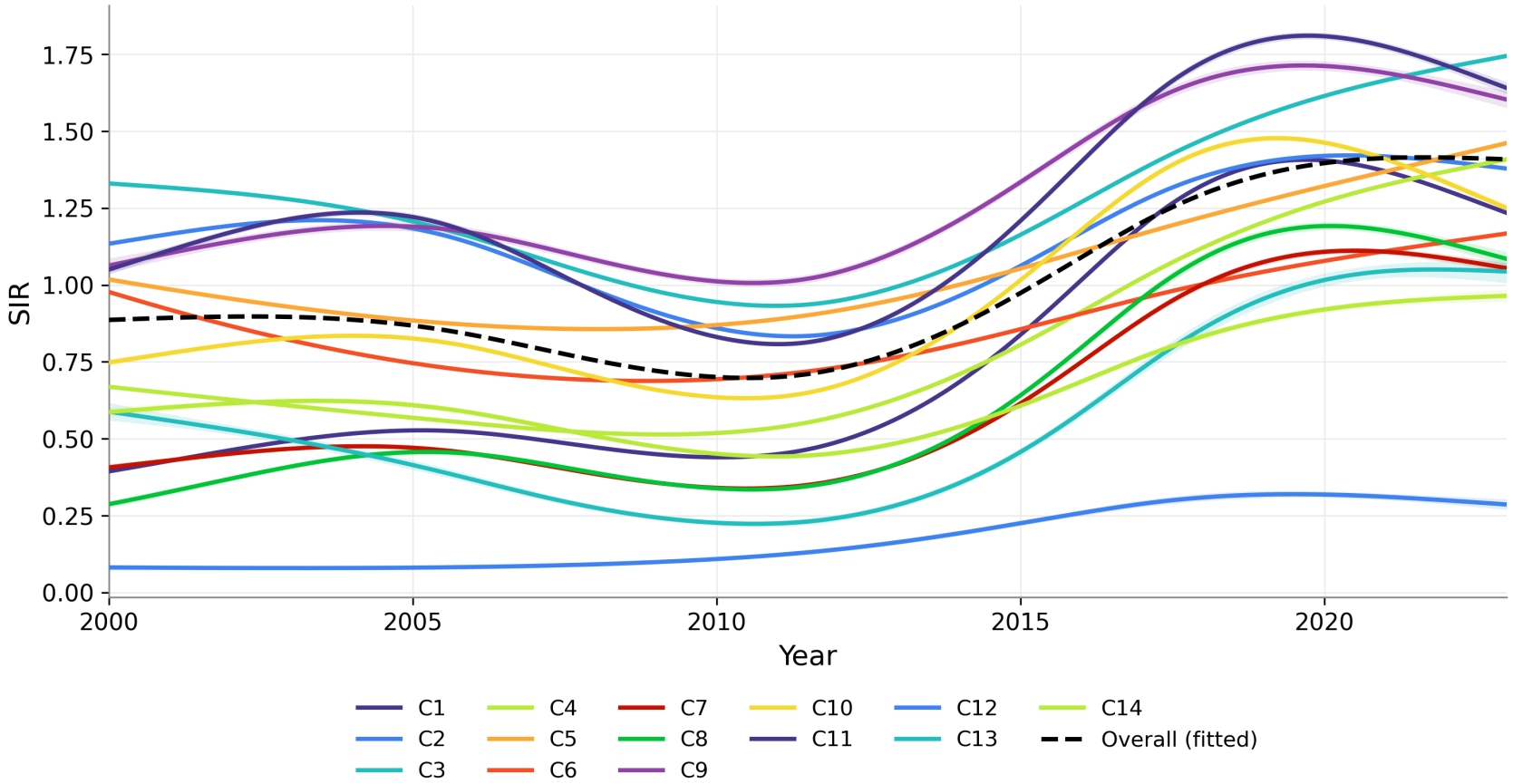
MSVTT: Cluster Map

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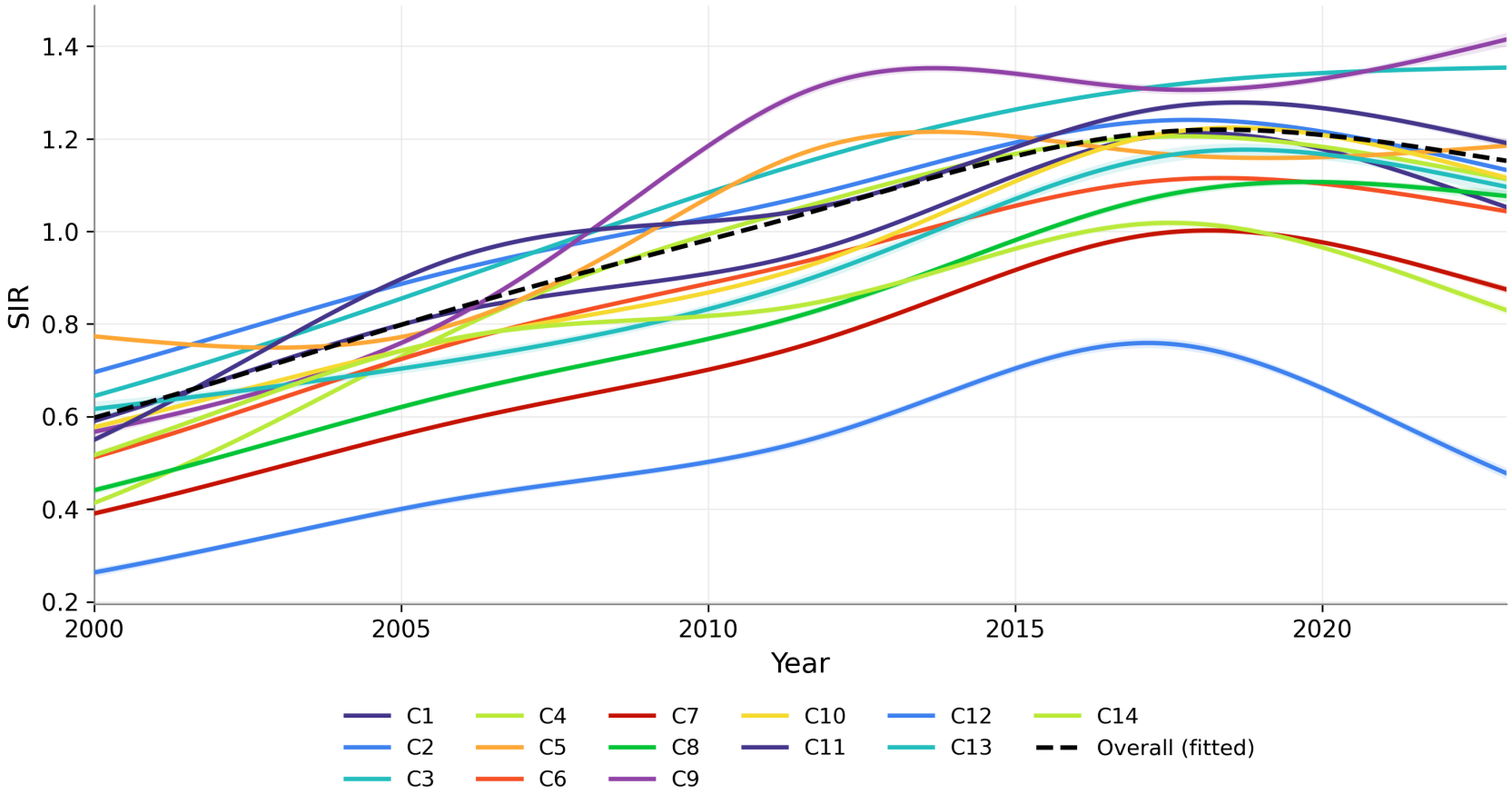
Gonorrhea: Fitted Trends

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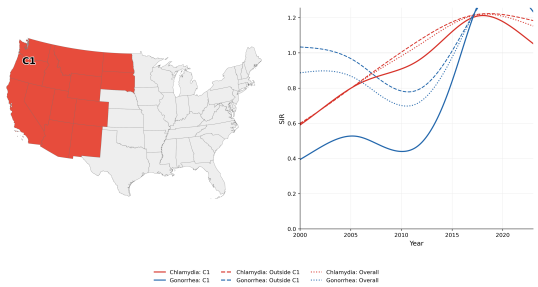
Shading: 95% pointwise model confidence intervals; cluster selection is not adjusted for.

Chlamydia: Fitted Trends

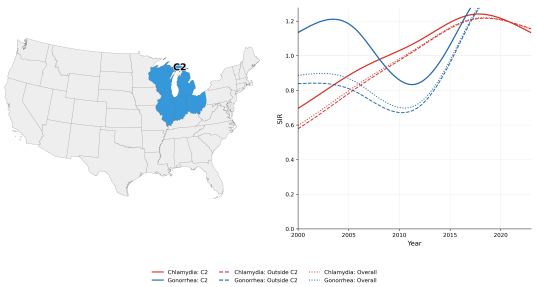


Shading: 95% pointwise model confidence intervals; cluster selection is not adjusted for.

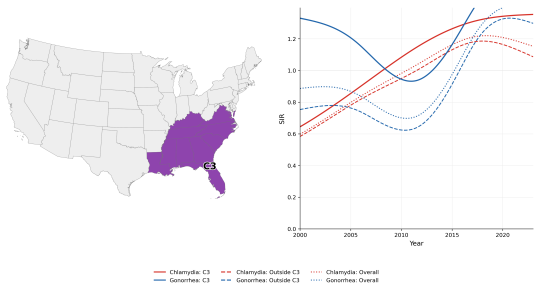
Detected clusters: 1–6



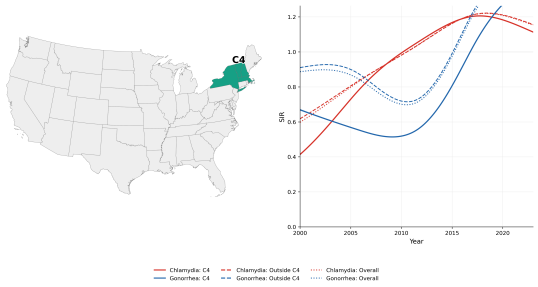
Cluster 1



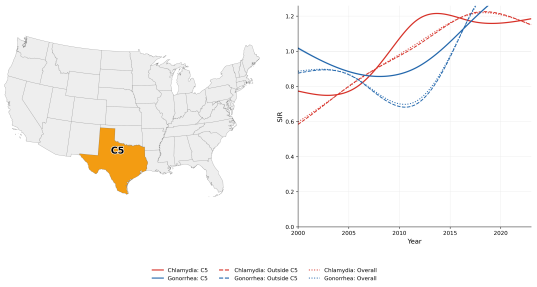
Cluster 2



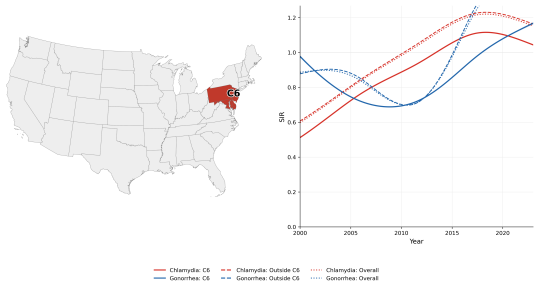
Cluster 3



Cluster 4

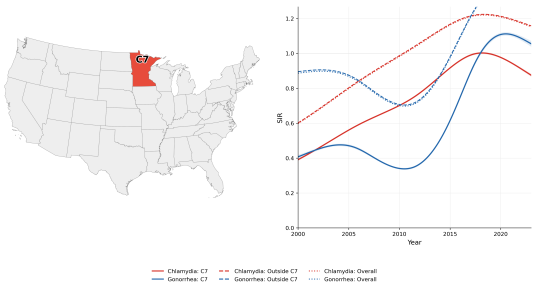


Cluster 5

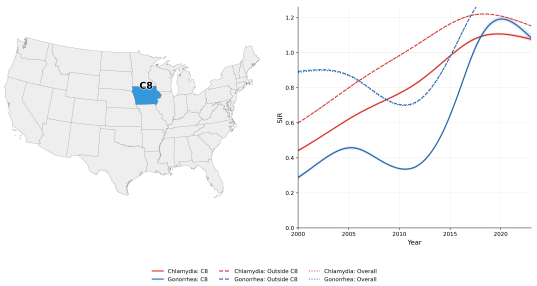


Cluster 6

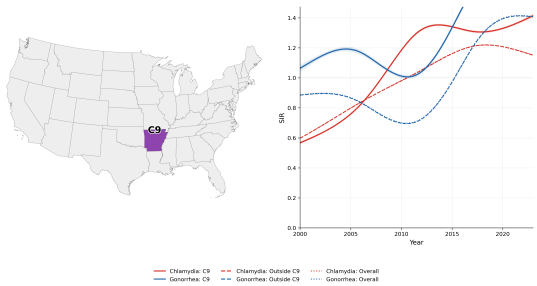
Detected clusters: 7–12



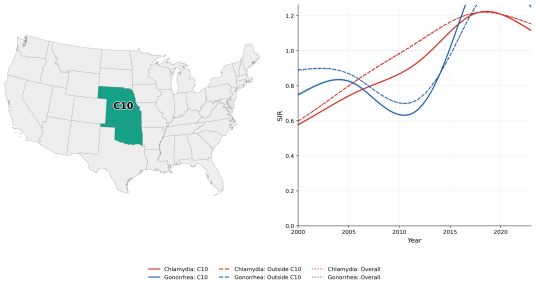
Cluster 7



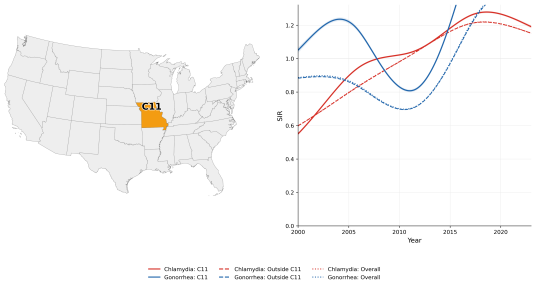
Cluster 8



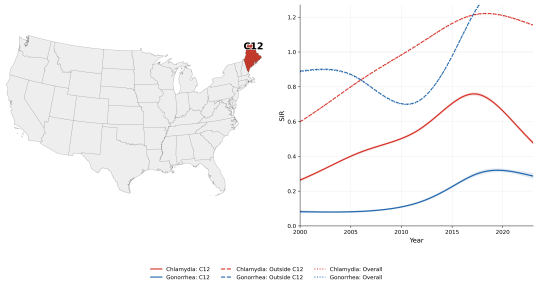
Cluster 9



Cluster 10

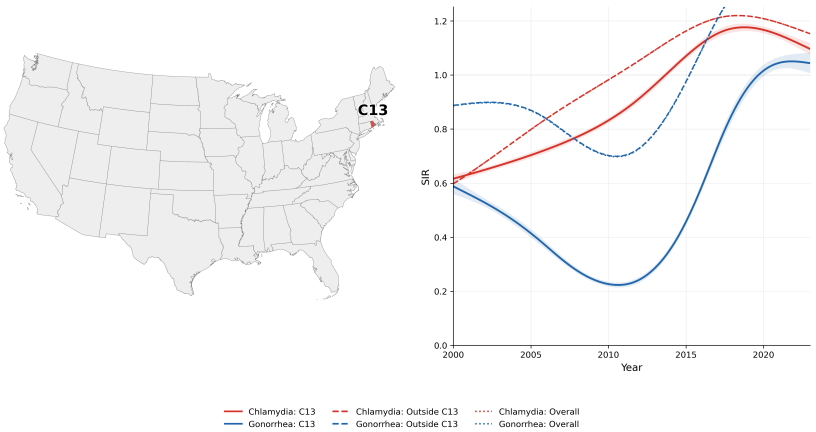


Cluster 11

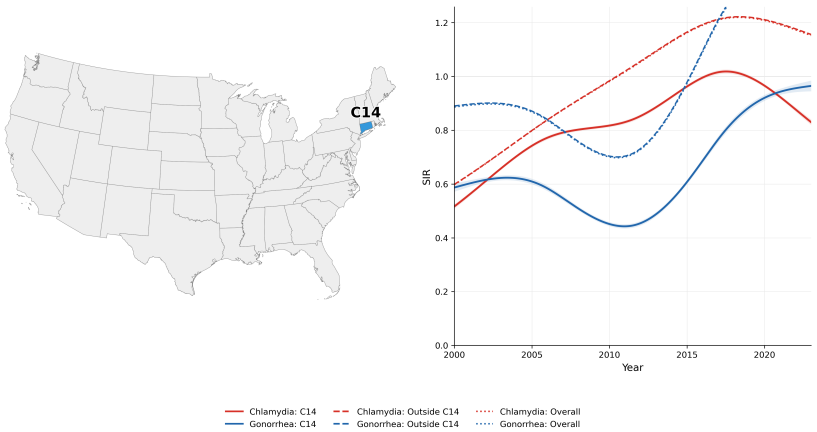


Cluster 12

Detected clusters: 13–14



Cluster 13



Cluster 14

Gonorrhea: Cluster Summary, 2000–2023

Gonorrhea clusters detected using multivariate SVTT with spline4.

Cluster	No. of states	Population person-years	Average annual population	Observed cases	Expected cases	SIR (O/E)	Incidence rate per 100,000	Disease LLR	Joint LLR	Joint *p*-value
C1	13	1,730,545,285	72,106,054	1,900,498	2,399,141.58	0.7922	109.821	68,136.0667	72,468.1739	0.0050
C2	5	1,115,356,569	46,473,190	1,769,683	1,546,274.66	1.1445	158.665	15,753.1010	22,162.2322	0.0050
C3	10	1,784,772,989	74,365,541	3,154,537	2,474,320.16	1.2749	176.747	17,446.3843	20,923.6535	0.0050
C4	4	674,867,632	28,119,485	753,641	935,602.79	0.8055	111.672	7,179.0580	17,175.9042	0.0050
C5	1	618,382,047	25,765,919	913,091	857,294.00	1.0651	147.658	4,224.7873	11,474.8532	0.0050
C6	5	691,857,159	28,827,382	836,991	959,156.22	0.8726	120.977	4,883.9863	5,131.9441	0.0050
C7	1	128,585,670	5,357,736	113,208	178,264.75	0.6351	88.041	2,231.5057	3,142.4829	0.0050
C8	1	73,447,506	3,060,313	65,081	101,823.96	0.6392	88.609	2,397.4849	3,139.6530	0.0050
C9	1	69,617,768	2,900,740	125,637	96,514.60	1.3017	180.467	347.9567	1,721.9757	0.0050
C10	3	202,568,576	8,440,357	273,961	280,830.96	0.9755	135.244	900.8501	1,672.1845	0.0050
C11	1	143,247,093	5,968,629	250,861	198,590.62	1.2632	175.125	456.7722	973.2085	0.0050
C12	1	31,872,297	1,328,012	7,792	44,186.16	0.1763	24.448	478.6845	928.0196	0.0050
C13	1	25,574,754	1,065,615	19,790	35,455.56	0.5582	77.381	710.4048	854.3163	0.0050
C14	1	84,979,278	3,540,803	77,260	117,811.03	0.6558	90.916	59.2832	684.3135	0.0050

Note: SIR = standardized incidence ratio; LLR = log-likelihood ratio. Incidence rates are per 100,000 person-years. Joint LLRs and *p*-values refer to the joint analysis of both diseases.

Chlamydia: Cluster Summary, 2000–2023

Chlamydia clusters detected using multivariate SVTT with spline4.

Cluster	No. of states	Population person-years	Average annual population	Observed cases	Expected cases	SIR (O/E)	Incidence rate per 100,000	Disease LLR	Joint LLR	Joint *p*-value
C1	13	1,730,545,285	72,106,054	7,066,141	7,312,675.22	0.9663	408.319	4,332.1072	72,468.1739	0.0050
C2	5	1,115,356,569	46,473,190	4,895,148	4,713,104.25	1.0386	438.886	6,409.1312	22,162.2322	0.0050
C3	10	1,784,772,989	74,365,541	8,322,693	7,541,822.40	1.1035	466.317	3,477.2692	20,923.6535	0.0050
C4	4	674,867,632	28,119,485	2,719,759	2,851,753.06	0.9537	403.006	9,996.8463	17,175.9042	0.0050
C5	1	618,382,047	25,765,919	2,721,885	2,613,064.86	1.0416	440.162	7,250.0659	11,474.8532	0.0050
C6	5	691,857,159	28,827,382	2,632,264	2,923,544.82	0.9004	380.464	247.9577	5,131.9441	0.0050
C7	1	128,585,670	5,357,736	408,876	543,357.78	0.7525	317.979	910.9773	3,142.4829	0.0050
C8	1	73,447,506	3,060,313	258,635	310,363.31	0.8333	352.136	742.1681	3,139.6530	0.0050
C9	1	69,617,768	2,900,740	325,650	294,180.18	1.1070	467.769	1,374.0190	1,721.9757	0.0050
C10	3	202,568,576	8,440,357	810,618	855,983.50	0.9470	400.170	771.3344	1,672.1845	0.0050
C11	1	143,247,093	5,968,629	628,479	605,311.79	1.0383	438.738	516.4363	973.2085	0.0050
C12	1	31,872,297	1,328,012	70,971	134,681.11	0.5270	222.673	449.3351	928.0196	0.0050
C13	1	25,574,754	1,065,615	98,317	108,069.91	0.9098	384.430	143.9114	854.3163	0.0050
C14	1	84,979,278	3,540,803	297,890	359,092.52	0.8296	350.544	625.0304	684.3135	0.0050

Note: SIR = standardized incidence ratio; LLR = log-likelihood ratio. Incidence rates are per 100,000 person-years. Joint LLRs and *p*-values refer to the joint analysis of both diseases.

Conclusions

Main Findings and Further Work



Main findings

Simulations and application illustrate MSVTT's ability to detect spatial differences in nonlinear disease risk trends while accounting for population sizes

Current limitations

Circular candidate windows, conditional independence across diseases, and computational cost

Next steps

- **Noncircular spatial clusters:** Extend the scan to connected regions with flexible shapes
- **Faster computation:** Optimize candidate scanning, reuse model calculations, and parallelize Monte Carlo replicates

Thank You



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Appendix

Quadratic SVTT Model



For each candidate cluster Z, let $g \in \{\text{in}, \text{out}\}$.

The aggregated observed counts are modeled as

$$Y_t^g(Z) \sim \text{Poisson}\left(\mu_t^g(Z)\right). \tag{1}$$

The quadratic temporal model is

$$\log \mu_t^g(Z) = \underbrace{\log m_t^g(Z)}_{\text{offset}} + \alpha_g + \beta_{1,g} x_t + \beta_{2,g} x_t^2. \tag{2}$$

Therefore, the relative-risk trajectory is

$$\text{RR}_t^g(Z) = \frac{\mu_t^g(Z)}{m_t^g(Z)} = \exp\left(\alpha_g + \beta_{1,g} x_t + \beta_{2,g} x_t^2\right). \tag{3}$$

- α_g: baseline log-relative risk;
- $\beta_{1,g}$: linear temporal trend;
- $\beta_{2,g}$: quadratic curvature.

Quadratic SVTT Hypotheses



For each candidate cluster Z, SVTT compares the temporal trend inside the cluster with the trend outside it.

Null hypothesis: common temporal trend

$$H_0(Z): \qquad \beta_{1,\text{in}} = \beta_{1,\text{out}}, \qquad \beta_{2,\text{in}} = \beta_{2,\text{out}}. \tag{4}$$

The inside and outside regions have the same linear effect and quadratic curvature.

Alternative hypothesis: different temporal trends

$$H_1(Z): \qquad \beta_{1,\text{in}} \neq \beta_{1,\text{out}} \quad \text{or} \quad \beta_{2,\text{in}} \neq \beta_{2,\text{out}}. \tag{5}$$

The trajectories differ through the linear component, quadratic curvature, or both.

Why are separate intercepts retained?

The intercepts α_{in} and α_{out} remain separate under H_0. This allows baseline risk levels to differ while the test focuses specifically on differences in temporal trends.

Poisson Likelihood Estimation



For $h \in \{0, 1\}$:

- $h = 0$: null model;
- $h = 1$: alternative model.

For candidate cluster Z, the Poisson log-likelihood is

$$\ell_h(Z; \boldsymbol{\theta}_h) = \sum_{g \in \{\text{in}, \text{out}\}} \sum_{t=1}^{T} \Big[Y_t^g(Z) \log \mu_{t,h}^g(Z) - \mu_{t,h}^g(Z) - \log\{Y_t^g(Z)!\} \Big].$$

The parameters are estimated separately under the two models:

$$\widehat{\boldsymbol{\theta}}_h(Z) = \arg\max_{\boldsymbol{\theta}_h} \ell_h(Z; \boldsymbol{\theta}_h), \qquad h \in \{0, 1\}.$$

The maximized log-likelihoods are

$$\ell_0(Z) = \ell_0\left(Z; \widehat{\boldsymbol{\theta}}_0\right), \qquad \ell_1(Z) = \ell_1\left(Z; \widehat{\boldsymbol{\theta}}_1\right).$$

Likelihood-Ratio Scan Statistic



For each candidate spatial window Z, the log-likelihood ratio is

$$\mathrm{LLR}(Z) = \ell_1(Z) - \ell_0(Z).$$

A large $\mathrm{LLR}(Z)$ indicates stronger evidence that the temporal trends inside and outside Z differ.

Observed scan statistic

$$T_{\mathrm{obs}} = \max_{Z \in \mathcal{Z}} \mathrm{LLR}(Z),$$

where $\mathcal{Z}$ is the collection of candidate spatial windows.

The most likely spatial cluster is

$$\widehat{Z} = \arg\max_{Z \in \mathcal{Z}} \ \mathrm{LLR}(Z).$$

Interpretation

$\widehat{Z}$ is the candidate window showing the strongest difference between its temporal trend and the trend outside the window.

Monte Carlo Significance

A conditional Monte Carlo bootstrap is used because the scan evaluates many strongly dependent spatial windows.

Global null model

$$\log\left\{\mu_{it}^{(0)}\right\} = \log(m_{it}) + \gamma_i + f_0(t).$$

- m_{it}: expected-count offset;
- γ_i: spatial-unit-specific baseline log-risk;
- $f_0(t)$: common temporal trajectory.

Parameter estimation

The parameters γ_i and $f_0(t)$ are estimated jointly by maximizing the Poisson likelihood.
An identifiability constraint is imposed, such as

$$\sum_{i=1}^{n} \gamma_i = 0.$$

Interpretation

- Baseline risk may differ across spatial units.
- All spatial units share the same temporal trajectory.
- Spatial variation in temporal trends is excluded under the null.

Monte Carlo Significance



Conditional on the observed total number of cases at time t, generate

$$\left(Y_{1t}^{(b)}, \ldots, Y_{nt}^{(b)}\right) \sim \text{Multinomial}\left(Y_{\cdot t}; \widehat{p}_{1t}^{(0)}, \ldots, \widehat{p}_{nt}^{(0)}\right),$$

where

$$Y_{\cdot t} = \sum_{i=1}^{n} Y_{it}, \qquad \widehat{p}_{it}^{(0)} = \frac{m_{it} \exp(\widehat{\gamma}_i)}{\sum_{j=1}^{n} m_{jt} \exp\left(\widehat{\gamma}_j\right)}.$$

The common temporal function $f_0(t)$ cancels from these conditional probabilities.

For every simulated dataset, repeat the complete spatial scan:

$$T_b = \max_{Z \in \mathcal{Z}} \text{LLR}^{(b)}(Z).$$

The Monte Carlo p-value is

$$\widehat{p} = \frac{1 + \sum_{b=1}^{B} \mathbb{I}\left(T_b \geq T_{\text{obs}}\right)}{B + 1}.$$

The detected cluster is significant when $\widehat{p} \leq \alpha$.

Restricted Cubic Splines: Basis and Knots



Restricted cubic splines represent nonlinear temporal trends using smooth piecewise cubic functions, with linearity imposed beyond the boundary knots [3].

Basis representation

Let x_t denote centered and scaled time. The K-dimensional basis is

$$\boldsymbol{B}_K(x) = [x, h_0(x), h_1(x), \ldots, h_{K-2}(x)]^\top .$$

Equivalently,

$$\boldsymbol{B}_K(x) = [B_1(x), \ldots, B_K(x)]^\top .$$

The temporal function is

$$f_K(t) = \boldsymbol{\theta}^\top \boldsymbol{B}_K(x_t) = \sum_{k=1}^{K} \theta_k B_k(x_t).$$

Internal knots

The $K-1$ internal knots are placed at equally spaced empirical quantiles:

$$q_j = \frac{j}{K}, \qquad \xi_j = Q_{j/K}(x), \qquad j = 1, \ldots, K-1.$$

The complete knot sequence is

$$\xi_0 < \xi_1 < \cdots < \xi_{K-1} < \xi_K.$$

Boundary knots

$$\xi_0 = \min_t x_t, \qquad \xi_K = \max_t x_t.$$

The spline may be nonlinear between ξ_0 and ξ_K but is constrained to be linear beyond these knots.

Construction and Properties



Truncated-cubic construction

Using the notation

$$(u)_+^3 = \max(u, 0)^3,$$

the nonlinear spline terms are

$$h_j(x) = (x - \xi_j)_+^3 - \frac{\xi_K - \xi_j}{\xi_K - \xi_{K-1}}(x - \xi_{K-1})_+^3$$

$$+ \frac{\xi_{K-1} - \xi_j}{\xi_K - \xi_{K-1}}(x - \xi_K)_+^3,$$

where

$$j = 0, \ldots, K - 2.$$

The correction terms impose the restricted, or linear-tail, condition.

Main properties

- Piecewise cubic between knots;
- smooth at every knot;
- linear beyond the boundary knots;
- K controls temporal flexibility.

Numerical implementation

- Basis columns are centered;
- columns are orthogonalized for numerical stability;
- the represented function space remains unchanged.

Spline SVTT Model



For each candidate spatial window Z, let $g \in \{\text{in}, \text{out}\}$.

Poisson count model

$$Y_t^g(Z) \sim \text{Poisson}\left(\mu_t^g(Z)\right).$$

Using the previously defined RCS basis,

$$f_{K,g}(t) = \boldsymbol{\theta}_g^\top \boldsymbol{B}_K(x_t) = \sum_{k=1}^{K} \theta_{k,g} B_k(x_t).$$

The Spline SVTT model is

$$\log \mu_t^g(Z) = \log m_t^g(Z) + \alpha_g + f_{K,g}(t).$$

Relative-risk trajectory

$$\text{RR}_t^g(Z) = \frac{\mu_t^g(Z)}{m_t^g(Z)} = \exp\left\{\alpha_g + \boldsymbol{\theta}_g^\top \boldsymbol{B}_K(x_t)\right\}.$$

Interpretation

- α_g controls the baseline log-relative risk.
- $\boldsymbol{\theta}_g$ controls the shape of the temporal trajectory.
- The RCS permits nonlinear trends with multiple changes in curvature.

Specifications considered

- Spline3: $K = 3$ basis functions;
- Spline4: $K = 4$ basis functions.

Hypotheses and Scan Statistic

Spline SVTT tests whether the temporal trajectory inside a candidate window differs from the trajectory outside it.

Null hypothesis

The inside and outside regions share the same spline coefficients:

$$H_0(Z) : \boldsymbol{\theta}_{\text{in}} = \boldsymbol{\theta}_{\text{out}}.$$

Equivalently,

$$\theta_{k,\text{in}} = \theta_{k,\text{out}}, \qquad k = 1, \ldots, K.$$

Separate intercepts are retained:

$$\alpha_{\text{in}} \neq \alpha_{\text{out}} \quad \text{is allowed.}$$

Thus, baseline risks may differ while the temporal shape remains common.

Alternative hypothesis

$$H_1(Z) : \theta_{k,\text{in}} \neq \theta_{k,\text{out}}$$

for at least one k.

The window-specific statistic is

$$\text{LLR}(Z) = \ell_1(Z) - \ell_0(Z).$$

The observed scan statistic is

$$T_{\text{obs}} = \max_{Z \in \mathcal{Z}} \text{LLR}(Z).$$

The same spatial scanning and Monte Carlo significance procedure described previously is then applied.

Methods Compared: NPFSS and URBFSS



NPFSS: Functional rank approach

- Extends the Wilcoxon–Mann–Whitney test to functional observations.
- Compares the distributions of complete trajectories inside and outside each candidate window.
- Uses pairwise functional signs instead of assuming Gaussian data.
- Robust and applicable to non-Gaussian functional observations.
- More suitable for differences extending across the temporal domain than for highly localized departures.

[4]

URBFSS: Pointwise rank approach

- Ranks observations across all spatial locations at each time point.
- Uses a standardized Wilcoxon rank-sum statistic to compare the inside and outside regions.
- Takes the maximum absolute rank statistic over time.
- Suitable for detecting temporally localized departures.
- Rank-based construction provides robustness to non-Gaussian observations and extreme values.

[1]

Methods Compared: PFSS and DFFSS



PFSS: Functional ANOVA approach

- Compares the mean functions inside and outside each candidate window.
- Measures between-group variation relative to variation among curves within the two groups.
- Integrates the difference over the complete observation period.
- Most suitable for differences that persist over a substantial part of the temporal domain.
- Performance may deteriorate for strongly non-Gaussian data or differences limited to a short interval.

[5]

DFFSS: Pointwise difference approach

- Calculates a standardized difference between the mean trajectories inside and outside each window.
- Evaluates this difference separately at each time point.
- Uses the largest standardized difference over time as the scan statistic.
- Suitable for detecting differences localized to only a few time points.
- Does not require a specified distribution, but assumes a common pointwise variance across locations.

[5]

Performance Measures

Statistical detection
Let R_0 and R_1 denote the numbers of null and non-null simulation replicates. Let $D_r = 1$ if at least one significant cluster is detected.

Type I error

$$\widehat{\text{Type I}} = \frac{1}{R_0} \sum_{r=1}^{R_0} D_r.$$

Power

$$\widehat{\text{Power}} = \frac{1}{R_1} \sum_{r=1}^{R_1} D_r.$$

Statistical significance is evaluated at

$$\alpha = 0.05.$$

Spatial recovery
Let $C^\star$ and $\widehat{C}$ denote the true and detected clusters. Define

$$TP = |C^\star \cap \widehat{C}|, \quad FN = |C^\star \setminus \widehat{C}|,$$
$$FP = |\widehat{C} \setminus C^\star|, \quad TN = n - TP - FN - FP.$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \qquad \text{Specificity} = \frac{TN}{TN + F}$$

$$\text{Precision} = \frac{TP}{TP + FP}.$$

$$\text{Jaccard} = \frac{|C^\star \cap \widehat{C}|}{|C^\star \cup \widehat{C}|} = \frac{TP}{TP + FP + FN}.$$

Two-Cluster Recovery Measures

Let $C_1^\star$, $C_2^\star$ be the true clusters and $\widehat{C}_1$, $\widehat{C}_2$ the detected clusters.

Optimal cluster matching

Pairwise Jaccard similarities:

$$J_{kj} = \frac{|C_k^\star \cap \widehat{C}_j|}{|C_k^\star \cup \widehat{C}_j|}, \qquad k, j \in \{1, 2\}.$$

Optimal matching:

$$\pi^\star = \arg\max_{\pi \in S_2} \sum_{k=1}^{2} J_{k,\pi(k)}.$$

Matched Jaccard index:

$$J_{\text{matched}} = \frac{1}{2} \sum_{k=1}^{2} J_{k,\pi^\star(k)}.$$

Both-cluster indicator:

$$I_{\text{both}}^{(r)} = \mathbb{I}\left(J_{1,\pi^\star(1)}^{(r)} > 0,\ J_{2,\pi^\star(2)}^{(r)} > 0\right).$$

Both-cluster detection rate:

$$\widehat{P}_{\text{both}} = \frac{1}{R} \sum_{r=1}^{R} I_{\text{both}}^{(r)}.$$

Partition-based measures

Let n_{kj} be the number of units from true class k assigned to detected class j, including the outside-cluster class.

Cluster purity:

$$\text{Purity} = \frac{1}{n} \sum_{j} \max_{k} n_{kj}.$$

Define the row and column totals:

$$a_k = \sum_{j} n_{kj}, \qquad b_j = \sum_{k} n_{kj}.$$

The expected agreement is

$$E = \frac{\left\{\sum_k \binom{a_k}{2}\right\}\left\{\sum_j \binom{b_j}{2}\right\}}{\binom{n}{2}}.$$

The adjusted Rand index is

$$\text{ARI} = \frac{\sum_{k,j} \binom{n_{kj}}{2} - E}{\frac{1}{2}\left[\sum_k \binom{a_k}{2} + \sum_j \binom{b_j}{2}\right] - E}.$$